ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ 2 SET SPECIAL I **WINTER SEMESTER, 2024-2025**

DURATION: 25 MINUTES **FULL MARKS: 20**

MATH 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID: _____

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1. Given, 2 + 2 (CO2) (PO1)

$$S = \{\begin{bmatrix} x^4 \\ y^{12} \end{bmatrix} \mid \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2\}$$

This set represents all vectors that are scaled by a factor of 2 from the original vector space $\mathbb{R}^2$. **Is S a vector space?** If yes, justify your answer. If no, provide a counterexample to demonstrate why it is not.

Solution:

For S to be a vector space, it must satisfy the following properties:

1. For any two vectors $\mathbf{u}, \mathbf{v} \in S$, their linear combination $k_1\mathbf{u} + k_2\mathbf{v}$ must also be in S (where k_1 and k_2 are scalars).
2. It must contain the zero vector.

Consider a vector corresponding to $(x, y) = (1, 1)$, which gives

$$\begin{bmatrix} x^4 \\ y^{12} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \in S.$$

Now consider scalar multiplication by a negative scalar:

$$-1 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}.$$

We check whether this resulting vector belongs to S.
There are no real values of x and y such that

$$x^4 = -1 \quad \text{and} \quad y^{12} = -1,$$

since $x^4 \geq 0$ and $y^{12} \geq 0$ for all real x, y. Hence,

$$\begin{bmatrix} -1 \\ -1 \end{bmatrix} \notin S.$$

Although the zero vector $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ is contained in S (taking $x = 0$ and $y = 0$), all vectors in S lie in the first quadrant with non-negative components. Therefore, S is not closed under scalar multiplication.
Hence, the set S does not satisfy the requirements of a vector space, and it is **not a vector space**.
Alternative ways to show this:

- You can show that S only contains vectors with non-negative components, so negative scalar multiples are not in S.
- You could also show that closure under addition fails by considering sums that produce negative components.

Rubrics:

- 1 mark for pointing out that S is not a vector space.
- 1 mark for correct reasoning/explanation.
- 2 marks for providing a valid counterexample.

2. Given, 16 (CO2) (PO1)

$$A = \begin{bmatrix} 1 & 1 & 3 & 0 & 2 \\ 0 & 1 & 2 & 1 & 3 \\ 1 & 1 & 3 & 1 & 4 \\ 1 & 2 & 5 & 1 & 5 \end{bmatrix}$$

Answer the following questions (After reducing it to its Reduced Row Echelon Form (RREF)):

(a) What is the dimension of Matrix A?

(b) What is the rank of Matrix A?

(c) Identify the pivot and free columns of Matrix A.

(d) For all the four fundamental subspaces associated with Matrix A (Columnspace $C(A)$, Nullspace $N(A)$, Rowspace $C(A^T)$, Left Nullspace $N(A^T)$), state the following:

- Their dimensions.
- Bases of the subspaces.
- The subspaces they belong to (i.e., $\mathbb{R}^m$ or $\mathbb{R}^n$, or something else).
- Which two pairs are orthogonal to each other and justify by showing the dot product of their basis vectors.

Solution:

We start with the augmented matrix $[A|b]$ based on the given system:

$$A = \begin{bmatrix} 1 & 1 & 3 & 0 & 2 \\ 0 & 1 & 2 & 1 & 3 \\ 1 & 1 & 3 & 1 & 4 \\ 1 & 2 & 5 & 1 & 5 \end{bmatrix}$$

Step 1: Eliminate entries below the first pivot in column 1. Perform $R_3 \to R_3 - R_1$ and $R_4 \to R_4 - R_1$:

$$A = \begin{bmatrix} 1 & 1 & 3 & 0 & 2 \\ 0 & 1 & 2 & 1 & 3 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 1 & 2 & 1 & 3 \end{bmatrix}$$

Step 2: Eliminate entries below the second pivot in column 2. Perform $R_4 \to R_4 - R_2$:

$$A = \begin{bmatrix} 1 & 1 & 3 & 0 & 2 \\ 0 & 1 & 2 & 1 & 3 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 3: Eliminate the entry above the third pivot in column 4. Perform $R_1 \to R_1 - R_3$ and $R_2 \to R_2 - R_3$:

$$\begin{bmatrix} 1 & 1 & 3 & -1 & 0 \\ 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 4: Eliminate the entry above the second pivot in column 2. Perform $R_1 \to R_1 - R_2$ and $R_1 \to R_1 + R_3$:

$$\text{RREF}([A|b]) = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

From the RREF form we can see that there are 3 pivot columns (1, 2, and 4) and 2 free columns (3 and 5). So, the rank of matrix A is 3 (Answer to 2(b)). For clarity, we can highlight the pivot columns in red and the free columns in green, and the pivots have been boxed (Answer to 2(c)):

$$\text{RREF}([A]) = \begin{bmatrix} \boxed{1} & 0 & 1 & 0 & 1 \\ 0 & \boxed{1} & 2 & 0 & 1 \\ 0 & 0 & 0 & \boxed{1} & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Columnspace and Nullspace:

The bases of the columnspace of A are the original columns of A corresponding to the pivot columns (1, 2, and 4):

$$\text{Basis } C(A) = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

The dimension of the columnspace is 3, and it is a subspace of $\mathbb{R}^4$.

As for the bases of its Nullspace, $N(A)$, we can write down $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$

We set free variables as I and pivot variables as $-F$, and thus we get the bases of the Nullspace,

$$\text{Basis } N(A) = \left\{ \begin{bmatrix} -1 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

Again, the transpose A^T is a 5×4 matrix:

$$A^T = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 3 & 2 & 3 & 5 \\ 0 & 1 & 1 & 1 \\ 2 & 3 & 4 & 5 \end{bmatrix}$$

Step 1: Eliminate entries below the first pivot in column 1. Perform $R_2 \to R_2 - R_1$, $R_3 \to R_3 - 3R_1$, and $R_5 \to R_5 - 2R_1$:

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 2 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 3 & 2 & 3 \end{bmatrix}$$

Step 2: Eliminate entries below the second pivot in column 2. Perform $R_3 \to R_3 - R_2$, $R_4 \to R_4 - R_2$, and $R_5 \to R_5 - 3R_2$:

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 \end{bmatrix}$$

Step 3: Eliminate entries below the second pivot in column 2. Perform $R_5 \to R_5 - 2R_4$, $R_1 =$

$R_1 - R_4$ and swap $R_3 \leftrightarrow R_4$:

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 3: Finalize RREF by swapping $R_3 \leftrightarrow R_4$ and clearing above the third pivot ($R_1 \to R_1 - R_3$, $R_5 \to R_5 - R_3$):

$$\text{RREF}(A^T) = \begin{bmatrix} \boxed{1} & 0 & 0 & 1 \\ 0 & \boxed{1} & 0 & 1 \\ 0 & 0 & \boxed{1} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

From the RREF form, we see 3 pivot columns (1, 2, and 3) and 1 free column (4). Thus, $\text{rank}(A^T) = 3$.

The basis for the rowspace of A (which is the columnspace of A^T) consists of the original columns of A^T corresponding to the pivots:

$$\text{Basis } C(A^T) = \left\{ \begin{bmatrix} 1 \\ 1 \\ 3 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \\ 1 \\ 4 \end{bmatrix} \right\}$$

For, the Left Nullspace $N(A^T)$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$, placing free variable as I and pivot variables as $-F$, we get the basis of the Left Nullspace as:

$$\text{Basis } N(A^T) = \left\{ \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Answers:

(a)

The **Dimension** of A is 4×5.

(b)

The **Rank** of A is 3.

(c)

The **Pivot Columns** of A are: 1, 2, and 4. (highlighted in red)

The **Free Columns** of A are: 3, and 5. (highlighted in green)

$$\text{RREF}([A]) = \begin{bmatrix} \boxed{1} & 0 & 1 & 0 & 1 \\ 0 & \boxed{1} & 2 & 0 & 1 \\ 0 & 0 & 0 & \boxed{1} & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(d)

Refer to the table below for the answers regarding the four fundamental subspaces associated with Matrix A.

(e)

The two pairs of orthogonal/perpendicular subspaces are:

- Columnspace $C(A) \perp$ Left Nullspace $N(A^T)$:
$$\begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix} = (-1) + 0 + 0 + 1 = 0$$
(The first vector is a basis vector of $C(A)$ and the second vector is the only basis vector of $N(A^T)$.)

- Nullspace $N(A) \perp$ Rowspace $C(A^T)$:
$$\begin{bmatrix} -1 \\ -1 \\ 0 \\ 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 3 \\ 0 \\ 2 \end{bmatrix} = (-1) + (-1) + 0 + 0 + 2 = 0$$
(The first vector is a basis vector of $N(A)$ and the second vector is a basis vector of $C(A^T)$.)

Rubrics:

- 0.5 mark for correct dimension of A.
- 0.5 mark for correct rank of A.
- 1 mark for correctly identifying pivot and free columns.
- 8 marks (1 mark for bases, 0.5 for dimensions and 0.5 for subspaces for each subspace) for correctly stating the bases, dimensions, and subspaces for all four fundamental subspaces.
- 1 mark (0.5 mark each) for correctly identifying the orthogonal pairs and justifying with dot product calculations.
- 5 marks (2.5 marks each) for correct RREF calculations and steps.

	Columnspace $C(A)$	**Nullspace** $N(A)$
Bases	$\left\{\begin{bmatrix}1\\0\\1\\1\end{bmatrix}, \begin{bmatrix}1\\1\\1\\2\end{bmatrix}, \begin{bmatrix}0\\1\\1\\1\end{bmatrix}\right\}$	$\left\{\begin{bmatrix}-1\\-2\\1\\0\\0\end{bmatrix}, \begin{bmatrix}-1\\-1\\0\\-2\\1\end{bmatrix}\right\}$
Dim	3	2
Subsp.	$\mathbb{R}^4$	$\mathbb{R}^5$

	Rowspace $C(A^T)$	**Left Nullspace** $N(A^T)$
Bases	$\left\{\begin{bmatrix}1\\1\\3\\0\\2\end{bmatrix}, \begin{bmatrix}0\\1\\2\\1\\3\end{bmatrix}, \begin{bmatrix}1\\1\\3\\1\\4\end{bmatrix}\right\}$	$\left\{\begin{bmatrix}-1\\-1\\0\\1\end{bmatrix}\right\}$
Dim	3	1
Subsp.	$\mathbb{R}^5$	$\mathbb{R}^4$